RHEL 8 LVM based Installation

Q.1 Boot up a RHEL 8 virtual machine and perform a custom installation using the graphical installer, making sure to select 'LVM' as the partitioning scheme for the root (/) partition.

Ans- Step 1 – lsblk

Step 2- vgs

Step 3- lvs

```
[root@localhost ~]# lvcreate -L 500M -n musicdata cs
  Logical volume "musicdata" created.
[root@localhost ~]# mkfs.ext4 /dev/cs/musicdata
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 512000 1k blocks and 128016 inodes
Filesystem UUID: 8bf3b7ab-bf02-41de-a6ff-498a1b3a3a26
Superblock backups stored on blocks:
        8193, 24577, 40961, 57345, 73729, 204801, 221185, 401409

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@localhost ~]# mkdir /mnt/music
[root@localhost ~]# mount /dev/cs/musicdata /mnt/music
[root@localhost ~]# blkid /dev/cs/musicdata
/dev/cs/musicdata: UUID="8bf3b7ab-bf02-41de-a6ff-498a1b3a3a26" TYPE="ext4"
[root@localhost ~]# vi /etc/fstab
[root@localhost ~]# mount -a
mount: /mnt/music: can't find UUID=<your-uuid>.
mount: (hint) your fstab has been modified, but systemd still uses
       the old version; use 'systemctl daemon-reload' to reload.
[root@localhost ~]# df -h
Filesystem                Size  Used Avail Use% Mounted on
devtmpfs                  4.0M     0  4.0M   0% /dev
tmpfs                     871M     0  871M   0% /dev/shm
tmpfs                     349M  7.2M  342M   3% /run
/dev/mapper/cs-root        17G  5.9G   12G  35% /
/dev/sda1                 960M  349M  612M  37% /boot
tmpfs                     175M   92K  175M   1% /run/user/0
/dev/sr0                   11G   11G     0 100% /run/media/root/CentOS-Stream-9-BaseOS-x86_64
/dev/mapper/cs-musicdata  459M   14K  430M   1% /mnt/music
```

Q.2 Inside your RHEL 8 system, use the 'lsblk' and 'vgs' commands to verify that your root partition is managed by LVM and note down the Volume Group and Logical Volume names.

Ans- Step 1- lsblk Step 2- pvcreate /dev/sdb

```
[root@localhost ~]# xfs_growfs /
meta-data=/dev/mapper/cs-root    isize=512    agcount=4, agsize=1113856 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
data     =                       bsize=4096   blocks=4455424, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
[root@localhost ~]# resize2fs /dev/cs/root
resize2fs 1.46.5 (30-Dec-2021)
resize2fs: Bad magic number in super-block while trying to open /dev/cs/root
Couldn't find valid filesystem superblock.
[root@localhost ~]# df -h
Filesystem           Size  Used Avail Use% Mounted on
devtmpfs             4.0M     0  4.0M   0% /dev
tmpfs                871M     0  871M   0% /dev/shm
tmpfs                349M  7.2M  342M   3% /run
/dev/mapper/cs-root   17G  5.9G   12G  35% /
/dev/sda1            960M  349M  612M  37% /boot
tmpfs                175M   92K  175M   1% /run/user/0
/dev/sr0              11G   11G     0 100% /run/media/root/CentOS-Stream-9-BaseOS-x86_64
```

**Q.3 Add a new virtual disk to your RHEL 8 VM, then use 'pvcreate', 'vgextend', and 'lvextend' to increase the size of your existing root Logical Volume by at least 1GB, and resize the filesystem to use the new space.

Hint: Use 'df -h' before and after to confirm the size change.**

Ans- Step 1- vgs

Step 2- vgextend cs /dev/sdb

Step 3- Logical volume: lvs

Step 4- lvextend -L +1G /dev/cs/root

Step 5- Resize filesystem: xfs_growfs /

Step 6- Verify: df -h

```
[root@localhost ~]# vgs
  VG       #PV #LV #SN Attr   VSize   VFree
  cs         1   2   0 wz--n- <19.00g    0
  musicvg    1   1   0 wz--n-  <2.00g 1.65g
[root@localhost ~]# vgextend cs /dev/sdc
  Volume group "cs" successfully extended
[root@localhost ~]# lvs
  LV         VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root       cs      -wi-ao---- <17.00g
  swap       cs      -wi-ao----   2.00g
  playlistlv musicvg -wi-a----- 352.00m
[root@localhost ~]# lvextend -L +1G /dev/cs/root
  Insufficient free space: 256 extents needed, but only 255 available
[root@localhost ~]# xfs_groefs /
bash: xfs_groefs: command not found...
[root@localhost ~]# xfs_growfs /
meta-data=/dev/mapper/cs-root    isize=512    agcount=4, agsize=1113856 blks
         =                       sectsz=512   attr=2, projid32bit=1
         =                       crc=1        finobt=1, sparse=1, rmapbt=0
         =                       reflink=1    bigtime=1 inobtcount=1 nrext64=0
data     =                       bsize=4096   blocks=4455424, imaxpct=25
         =                       sunit=0      swidth=0 blks
naming   =version 2              bsize=4096   ascii-ci=0, ftype=1
log      =internal log           bsize=4096   blocks=16384, version=2
         =                       sectsz=512   sunit=0 blks, lazy-count=1
realtime =none                   extsz=4096   blocks=0, rtextents=0
```

```
root@localhost:~
[root@localhost ~]# lsblk
NAME                   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda                      8:0    0   20G  0 disk
├─sda1                   8:1    0    1G  0 part /boot
└─sda2                   8:2    0   19G  0 part
  ├─cs-root            253:0    0   17G  0 lvm  /
  └─cs-swap            253:1    0    2G  0 lvm  [SWAP]
sdb                      8:16   0    2G  0 disk
└─sdb1                   8:17   0    2G  0 part
  └─musicvg-playlistlv 253:2    0  352M  0 lvm
sdc                      8:32   0    1G  0 disk
└─cs-musicdata         253:3    0  500M  0 lvm  /mnt/music
sr0                     11:0    1 10.8G  0 rom  /run/media/root/CentOS-Stream-9-BaseOS-x86_64
```

**Q.4 Create a new Logical Volume called 'musicdata' of 500MB in your existing Volume Group, format it with ext4, mount it at /mnt/music, and add an entry to /etc/fstab so it mounts automatically on boot.

Constraint: Use only command-line tools (no GUI) for this task.**

Ans- Step 1- lvcreate -L 500M -n musicdata cs

Step 2- format: mkfs.ext4 /dev/cs/musicdara

Step 3 – Create mount point- mkdir /mnt/music

Step 4- Mount- mount /dev/cs/musicdata/mnt/music

Step 5- Get UUID- blkid /dev/cs/musicdata

Edit fstab: vi/etc/fstab

```
[root@localhost ~]# lsblk
NAME       MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda          8:0    0   20G  0 disk
├─sda1       8:1    0    1G  0 part /boot
└─sda2       8:2    0   19G  0 part
  ├─cs-root
           253:0    0   17G  0 lvm  /
  └─cs-swap
           253:1    0    2G  0 lvm  [SWAP]
sdb          8:16   0    2G  0 disk
└─sdb1       8:17   0    2G  0 part
  └─musicvg-playlistlv
           253:2    0  352M  0 lvm
sr0         11:0    1 10.8G  0 rom  /run/media/user/CentOS-Stream-9-BaseOS-x86_64
[root@localhost ~]# vgs
  VG      #PV #LV #SN Attr   VSize   VFree
  cs        1   2   0 wz--n- <19.00g    0
  musicvg   1   1   0 wz--n-  <2.00g 1.65g
[root@localhost ~]# lvs
  LV         VG      Attr       LSize   Pool Origin Data%  Meta%  Move Log Cpy%Sync Convert
  root       cs      -wi-ao---- <17.00g
  swap       cs      -wi-ao----   2.00g
  playlistlv musicvg -wi-a----- 352.00m
```